

Synthetic Stage One Technical Report

Selective intervention, authority allocation, distribution shift, and claim-bounded evidence

Core Research Document - Frozen evidence report

Human-AI Codecision Systems Research Program

Stage One revision - September 2026

Evidence status: Synthetic Stage One contains five independent seeds, 180 trials per policy and seed, 3,600 total decisions, and a controlled AI distribution shift in the final quarter. There were zero human participants. Adaptive codecision selectively regulated authority but did not improve synthetic accuracy or declared utility over human-only control. Human trust, satisfaction, and real cognitive state remain unmeasured.

Executive summary

The frozen Synthetic Stage One bundle contains five independent seeds, 180 trials per policy and seed, and 3,600 total decisions. The adaptive policy escalated 8.22% of trials, overrode 0.33%, and generated 0.11% unnecessary overrides. This validates selective intervention behavior. It did not improve accuracy or declared utility over human-only control. Aggressive automation achieved the highest mean synthetic accuracy by a very small margin but eliminated autonomy, overrode every trial, and generated 21.11% unnecessary overrides, yielding the lowest utility.

1. Aggregate results

Policy	Accuracy	Escalation	Override	Unnecessa ry	Autonomy	Utility
Human only	0.5589 ± .0606	0	0	0	1.0000	0.6186 ± .0253
Static recommendation	0.5611 ± .0580	0	0	0	0.7990	0.5834 ± .0246
Adaptive codecision	0.5600 ± .0575	0.0822 ± .0182	0.0033 ± .0050	0.0011 ± .0025	0.7701	0.5787 ± .0268
Aggressive automation	0.5633 ± .0554	1.0000	1.0000	0.2111 ± .0283	0.0000	0.3824 ± .0296

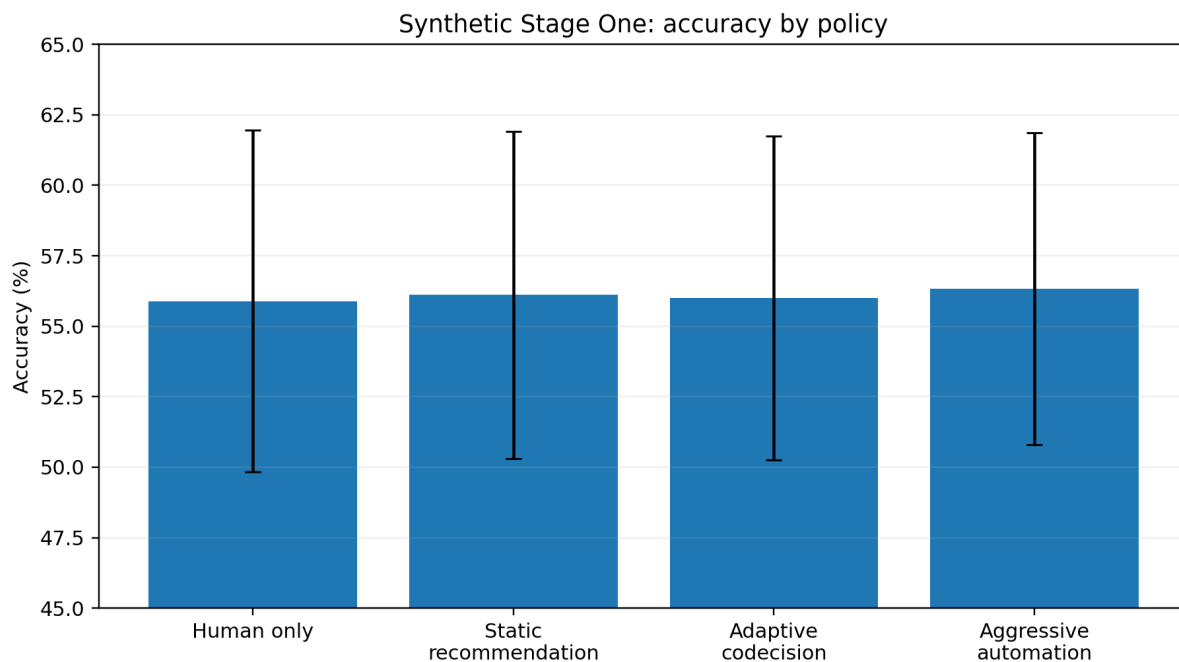


Figure 1. Measured mean synthetic accuracy ± sample SD.

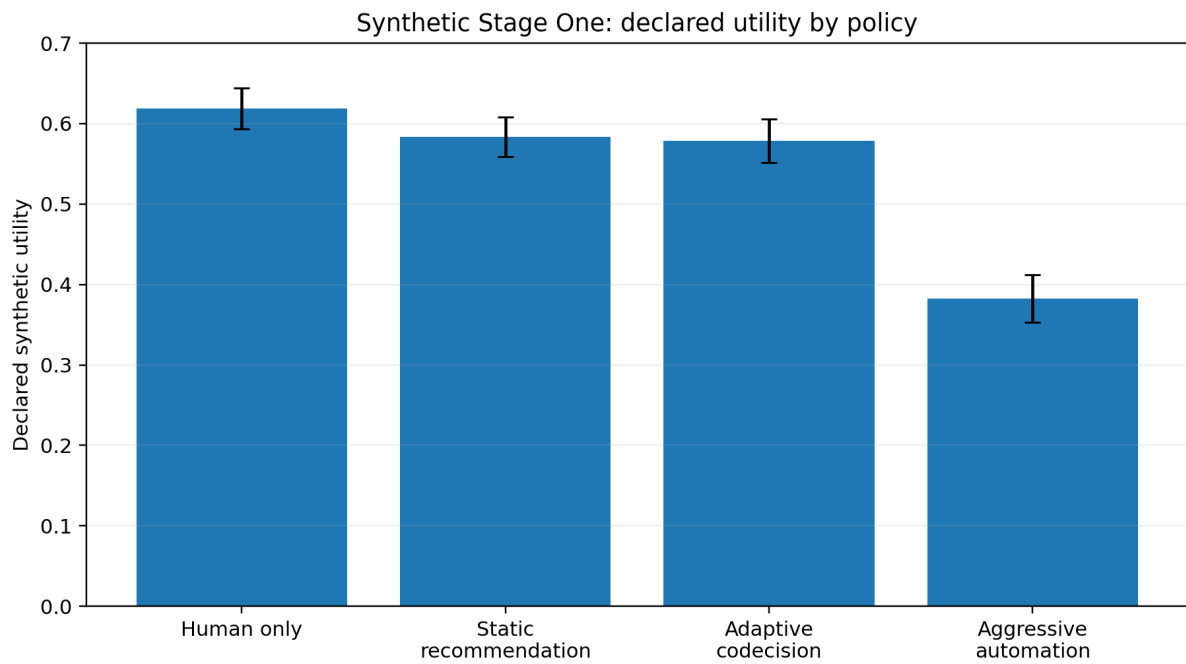


Figure 2. Measured declared synthetic utility $\pm$ sample SD.

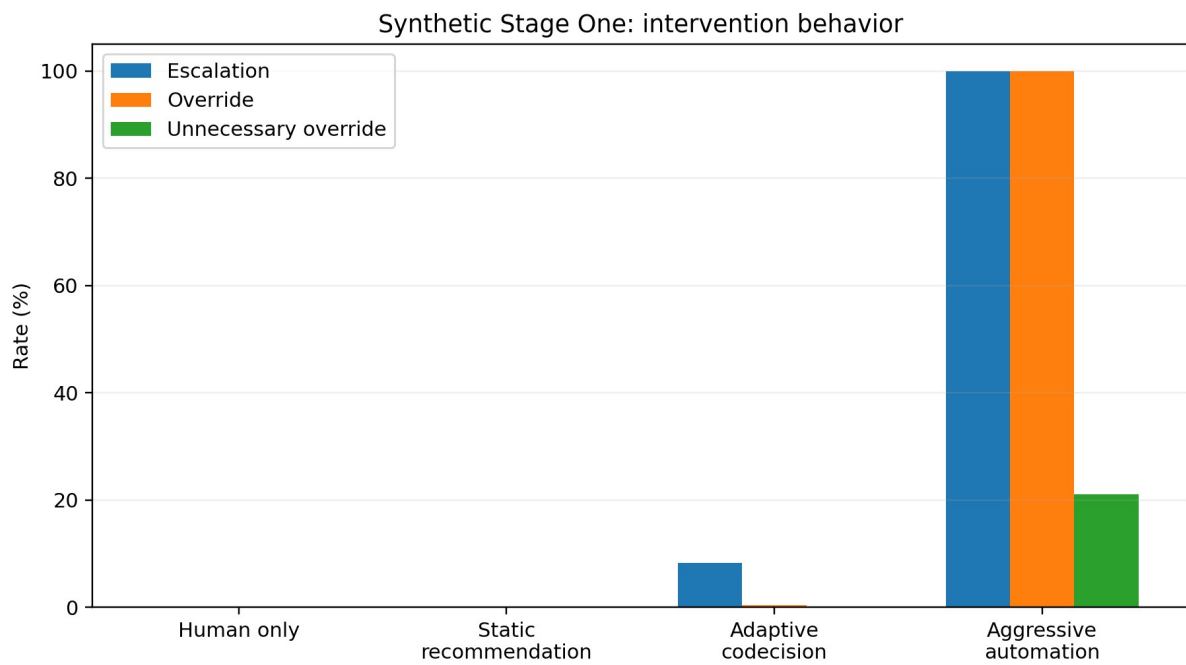


Figure 3. Measured escalation, override, and unnecessary-override rates.

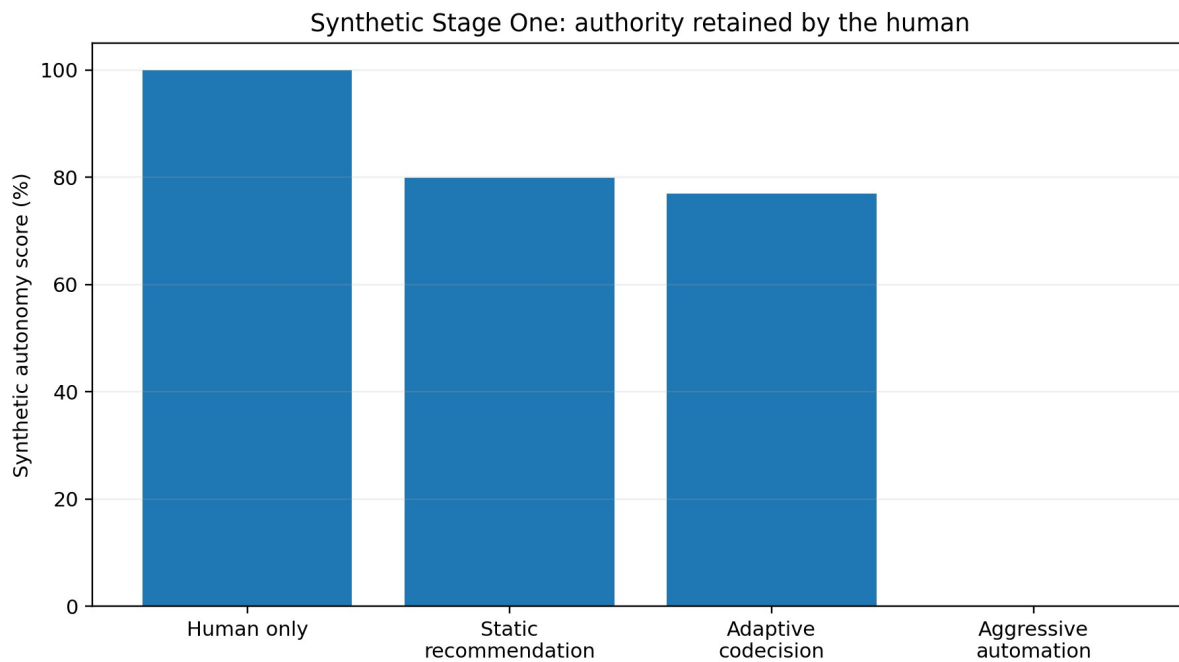


Figure 4. Synthetic authority/autonomy score by policy.

2. Interpretation

The adaptive policy's principal success is behavioral selectivity: unlike aggressive automation, it intervenes rarely and overrides exceptionally rarely. That selectivity does not translate into better accuracy or declared utility under the current configuration. The experiment therefore separates a mechanism-validation claim from an efficacy claim.

3. Threats to validity

- Synthetic human behavior is generated by a model and cannot validate real human uncertainty, trust, or workload.
- The declared utility embeds design choices about the relative value of accuracy, autonomy, and intervention costs.
- Five seeds provide preliminary replication but may not resolve small policy effects.
- Threshold defaults may be poorly matched to the simulated human/AI skill regime.
- The cybersecurity classification environment is intentionally simplified.
- Distribution shift is controlled and synthetic rather than evidence of robustness in deployed systems.

4. Next experiment

Run human-skill x AI-skill sweeps, threshold ablations, and held-out seeds before modifying the headline hypothesis. Preserve the frozen Stage One bundle as the untuned baseline.